245B, notes 0a. An alternate approach to the Carathéodory extension theorem

Terence Tao · 3 January 2009

https://terrytao.wordpress.com/2009/01/03/254a-notes-0a-an-alternate-approach-to-the-caratheodory-extension-theorem/

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In this supplemental note to the previous lecture notes, I would like to give an alternate proof of a (weak form of the) Carathéodory extension theorem. This argument is restricted to the σ-finite case, and does not extend the measure to quite as large a σ-algebra as is provided by the standard proof of this theorem, but I find it conceptually clearer (in particular, hewing quite closely to Littlewood's principles, and the general Lebesgue philosophy of treating sets of small measure as negligible), and suffices for many standard applications of this theorem, in particular the construction of Lebesgue measure.

Let us first state the precise statement of the theorem:

> **Theorem 1.** (Weak Carathéodory extension theorem) Let $\mathcal{A}$ be a Boolean algebra of subsets of a set X, and let $\mu : \mathcal{A} \to [0, +\infty]$ be a function obeying the following three properties:
>
> 1. $\mu(\emptyset) = 0$.
> 2. (Pre-countable additivity) If $A_1, A_2, \ldots \in \mathcal{A}$ are disjoint and such that $\bigcup_{n=1}^\infty A_n$ also lies in $\mathcal{A}$, then $\mu(\bigcup_{n=1}^\infty A_n) = \sum_{n=1}^\infty \mu(A_n)$.
> 3. (σ-finiteness) X can be covered by at most countably many sets in $\mathcal{A}$, each of which has finite μ-measure.
>
> Let $\mathcal{X}$ be the σ-algebra generated by $\mathcal{A}$. Then μ can be uniquely extended to a countably additive measure on $\mathcal{X}$.

We will refer to sets in $\mathcal{A}$ as *elementary sets* and sets in $\mathcal{X}$ as *measurable sets*. A typical example is when X=[0,1] and $\mathcal{A}$ is the collection of all sets that are unions of finitely many intervals; in this case, $\mathcal{X}$ are the Borel-measurable sets.

— Some basics —

Let us first observe that the hypotheses on the premeasure μ imply some other basic and useful properties:

1. From property 1. and 2. we see that μ is finitely additive (thus $\mu(A_1 \cup \ldots \cup A_n) = \mu(A_1) + \ldots + \mu(A_n)$ whenever $A_1, \ldots, A_n$ are disjoint elementary sets).
2. As particular consequences of finite additivity, we have monotonicity ($\mu(A) \leq \mu(B)$ whenever $A \subset B$ are elementary sets) and finite subadditivity ($\mu(A_1 \cup \ldots \cup A_n) \leq \mu(A_1) + \ldots + \mu(A_n)$ for all elementary $A_1, \ldots, A_n$, not necessarily disjoint).
3. We also have pre-countable subadditivity: $\mu(A) \leq \sum_{n=1}^\infty \mu(A_n)$ whenever the elementary sets $A_1, A_2, \ldots$ cover the elementary set A. To see this, first observe by replacing A_n with $A_n \setminus \bigcup_{i=1}^{n-1} A_i$ and using monotonicity that we may take the A_i to be disjoint; next, by restricting all the A_i to A and using monotonicity we may assume that A is the union of the A_i, and now the claim is immediate from pre-countable additivity.

— Existence —

Let us first verify existence. As is standard in measure-theoretic proofs for σ-finite spaces, we first handle the finite case (when $\mu(X) < \infty$), and then rely on countable additivity or sub-additivity to recover the σ-finite case.

The basic idea, following Littlewood's principles, is to view the measurable sets as lying in the "completion" of the elementary sets, or in other words to exploit the fact that measurable sets can be approximated to arbitrarily high accuracy by elementary sets.

Define the *outer measure* $\mu_*(A)$ of a set $A \subset X$ to be the infimum of $\sum_{n=1}^{\infty} \mu(A_n)$, where $A_1, A_2, \ldots$ range over all at most countable collections of elementary sets that cover A. It is clear that outer measure is monotone and countably subadditive. Also, since μ is pre-countably subadditive, we see that $\mu_*(A) \geq \mu(A)$ for all elementary A. Since we also have the trivial inequality $\mu_*(A) \leq \mu(A)$, we conclude that μ_* and μ agree on elementary sets.

The outer measure naturally defines a pseudometric (and thus a topology) on the space of subsets of X, with the distance between A and B being defined as $\mu_*(A \Delta B)$, where Δ denotes symmetric difference. (The subadditivity of μ_* ensures the triangle inequality; furthermore, we see that the Boolean operations (union, intersection, complement, etc.) are all continuous with respect to this pseudometric.) With this pseudometric, we claim that the measurable sets lie in the closure of the elementary sets. Indeed, it is not difficult to see (using subadditivity and monotonicity properties of μ_*) that the closure of the elementary sets are closed under finite unions, under complements, and under countable disjoint unions (here we need finiteness of $\mu(X)$ to keep the measure of all the pieces absolutely summable), and thus form a σ-algebra. Since this σ-algebra clearly contains the elementary sets, it must contain the measurable sets also.

By subadditivity of μ_*, the function $A \mapsto \mu_*(A)$ is Lipschitz continuous. Since this function is finitely additive on elementary sets, we see on taking limits (using subadditivity to control error terms) that it must be finitely additive on measurable sets also. Since μ_* is finitely additive, monotone, and countably sub-additive, it must be countably additive, and so μ_* is the desired extension of μ to the measurable sets. This completes the proof of the theorem in the finite measure case.

To handle the σ-finite case, we partition X into countably many elementary sets of finite measure, and use the above argument to extend μ to measurable subsets of each such elementary set. It is then a routine matter to sum together these localised measures to recover a measure on all measurable sets; the pre-countable additivity property ensures that this sum still agrees with μ on elementary sets.

— Uniqueness —

Now we verify uniqueness. Again, we begin with the finite measure case.

Suppose first that $\mu(X) < \infty$, and that we have two different extensions $\mu_1, \mu_2 : \mathcal{X} \to [0, +\infty]$ of μ to $\mathcal{X}$ that are countably additive. Observe that μ_1, μ_2 must both be continuous with respect to the μ_* pseudometric used in the existence argument, from countable subadditivity; since every measurable set is a limit of elementary sets in this pseudometric, we obtain uniqueness in the finite measure case.

When instead X is σ-finite, we cover X by elementary sets of finite measure. The previous argument shows that any two extensions μ_1, μ_2 of μ agree when restricted to each of these sets, and the claim then follows by countable additivity. □

Remark 1. The uniqueness claim fails when the σ-finiteness condition is dropped. Consider for instance the rational numbers $X = \mathbb{Q}$, and let the elementary sets be the finite unions of intervals $[a, b) \cap \mathbb{Q}$. Define the measure $\mu(A)$ of an elementary set to be zero if A is empty, and $+\infty$ otherwise. As the rationals are countable, we easily see that every set of rationals is measurable. One easily verifies the pre-countable additivity condition (though the σ-finiteness condition fails horribly). However, μ has multiple extensions to the measurable sets; for instance, any positive scalar multiple of counting measure is such an extension. ⋄

Remark 2. It is not difficult to show that the measure completion $\overline{\mathcal{X}}$ of $\mathcal{X}$ with respect to μ is the same as the topological closure of $\mathcal{X}$ (or of $\mathcal{A}$) with respect to the above pseudometric. Thus, for instance, a subset of [0,1] is Lebesgue measurable if and only if it can be approximated to arbitrary accuracy (with respect to outer measure) by a finite union of intervals. ⋄

A particularly simple case of Theorem 1 occurs when X is a metrisable compact Hausdorff totally disconnected space (i.e. a metrisable Stone space), such as the infinite discrete cube $\{0,1\}^{\mathbb{N}}$ or any other Cantor space. Then (see forthcoming lecture notes) the Borel σ-algebra $\mathcal{X}$ is generated by the Boolean algebra $\mathcal{A}$ of clopen sets. Also, as clopen sets here are simultaneously compact and open, we see that any infinite cover of one clopen set by others automatically has a finite subcover. From this, we conclude

> **Corollary 1.** Let X be a compact Hausdorff totally disconnected space. Then any finitely additive σ-finite measure on the clopen sets uniquely extends to a countably additive measure on the Borel sets.

By identifying $\{0,1\}^{\mathbb{N}}$ with $[0,1]$ up to a countable set, this provides one means to construct Lebesgue measure on $[0,1]$; similar constructions are available for $\mathbb{R}$ or $\mathbb{R}^n$.

Comments (23)



mmailliw/william · 3 January 2009

Just below "Existence" you refer to "the finite measure case mu(X) < 0"; the 0 should be an infinity (otherwise the measure is negative!)



Américo Tavares · 3 January 2009

Dear Professor Tao,
In the title: don't you want to say 245B, notes 0a?



Terence Tao · 3 January 2009

Thanks for the corrections!

JB · 4 January 2009

Hello,
Thank you for this post.
Two very small remarks :
– on wikipedia, "Boolean algebra" does not refer to "algebra over a set".
– the outer measure only defines a pseudometric.
[Corrected, thanks – T.]



Américo Tavares · 18 January 2009

Typo in the correction (in the title): 245B instead of 254B. *[Corrected, thanks – T.]*



实分析 0-10 « Liu Xiaochuan's Weblog · 16 February 2009

[...] 第零点五节是 Carathéodory 测度扩张定理的一个新的证明，而最终引出了一个 Lebesgue 测度的新的引入方式。我跟向开南教授聊起这个，他笑着说："其实也不简单，这个定理听起来好像更快一些，但是需要的知识却更多了。"我很有同感。在一个乘积空间上的乘积测度，然后再转换成[0,1]区间上的测度。无论怎样，这是一个挺有趣的角度。 [...]



Max Menzies · 2 November 2009

Dear Terry,
in the second paragraph under the heading 'uniqueness' you say that the set of measurable A on which mu_1 and mu_2 agree is a sigma-algebra: you say that it is closed under countable unions by the monotone convergence property of sets.
But this is not obvious to me, it is only obvious that this set is closed under disjoint unions (it's not obvious the set is even closed under the union of two sets). The only way I can see that this set is a sigma-algebra is to use a result that the smallest monotone class which contains some given algebra

of sets is a sigma-algebra (it IS clear that the set in question is a monotone class, by the monotone convergence properties of measures).



Terence Tao · 2 November 2009

Hmm, good point. Fortunately, I think the existence argument gives uniqueness for free anyway; I've reorganised the text accordingly.

משפט ההרחבה של Carathéodory מהומה רבה על לא דבר » ארכיון » What's new · 28 February 2010

[...] שגיליתי רק לאחרונה דרך פוסט בבלוג של טרנס טאו. גישה זו היא יותר גישה אחרת להוכחת משפט Carathéodory, אינטואיטיבית ופחות [...]



254B, Notes 2: Roth's theorem « What's new · 8 April 2010

[...] Define the outer measure of a subset of (not necessarily a limit subset) to be the infimum of , where is a countable family of limit subsets of that cover , and call a subset of null if it has zero outer measure. Call a subset Loeb measurable if it differs from a limit set by a null set. Show that there is a unique extension of Loeb measure from limit sets to Loeb measurable sets that is a countably additive probability measure on . (Hint: use the Carathéodory extension theorem, see e.g. my 254B notes 0 or notes 0a.) [...]



Michael Greinecker · 23 July 2010

It is actually possible to give a proof of the full extension theorem along similar lines. The details are in:
Maharam, Dorothy
From finite to countable additivity.
Port. Math. 44, 265-282 (1987).
http://purl.pt/3098

משפט ההרחבה של Carathéodory | מהומה רבה על לא דבר August 2010

[...] שגיליתי רק לאחרונה דרך פוסט בבלוג של טרנס טאו. גישה זו היא יותר גישה אחרת להוכחת משפט Carathéodory, אינטואיטיבית ופחות [...]



245A, Notes 1: Lebesgue measure « What's new · 9 September 2010

[...] For a further discussion of how measures can be viewed as completions of elementary measures, see these notes. [...]



G. Rodrigues · 23 April 2011

As a matter of terminology, what you call the Borel sigma-algebra should be called the Baire sigma-algebra. The Baire sigma-algebra is the one generated by the continuous functions = the one generated by the clopens if the space is compact, Hausdorff, totally-disconnected. The Borel sigma-algebra is the sigma-algebra generated by the open sets, which is usually larger than the Baire algebra.



Terence Tao · 23 April 2011

Hmm, fair enough. I restricted the relevant remark to the metrisable case, in which the Baire and Borel sigma algebras should be the same.



Murali · 12 October 2011

Can I have an example of a set that is part of Lebesgue sigma algebra but not part of borel sigma algebra and similarly for P(R)?



Terence Tao · 12 October 2011

1. See Exercise 16 of https://terrytao.wordpress.com/2010/09/25/245a-notes-3-integration-on-abstract-measure-spaces-and-the-convergence-theorems/
2. See Section 4 of https://terrytao.wordpress.com/2010/09/09/245a-notes-1-lebesgue-measure/



Boris Tsirelson · 23 February 2013

See arXiv:0712.2270, Jun Tanaka and Peter F. McLoughlin, "A realization of measurable sets as limit points".



Fabio Enrico Traverso · 9 July 2020

Hi! How does this less general result link to the construction of a lebesgue measure?
Thank you



Aditya Guha Roy · 23 March 2022

Reblogged this on Aditya Guha Roy's weblog.



Sam · 2 December 2025

Dear professor Tao:
Can you explain a little more about the statement that "the Boolean operations (union, intersection, complement, etc.) are all continuous with respect to this pseudometric"?
[Hint: what does it mean for addition of vectors to be continuous with respect to the Euclidean metric? -T.]



Anonymous · 5 December 2025

Dr Tao,
I also wonder why is it called "pseudo-" metric, also "pseudo-"randomness, "pesudo-" differential , and so forth.



Terence Tao · 6 December 2025

The mathematical usage of this prefix is similar to the general usage, though without the perjorative connotation: a "pseudo-X" is an object that exhibits many of the features of X, but is not necessarily an actual example of X. ("quasi-" is a similar prefix, although typically if both prefixes are in use, a "quasi-X" has a weaker resemblance to an actual "X" than a "pseudo-X". There is also "semi-", though a "semi-X" typically obeys some partial subset of the axioms defining X, rather than being a parallel object that merely resembles X. Finally, a "weak X" would actually be an example of an X, rather than only partially being of this type, but would typically have fewer desirable features than a "strong X" example of an X object.)
Incidentally, modern large language models are rather good at explaining these sorts of linguistic nuances.